Form A

1. (a) $P(A^c) = 1 - P(A) = 1 - 0.4 = 0.6$

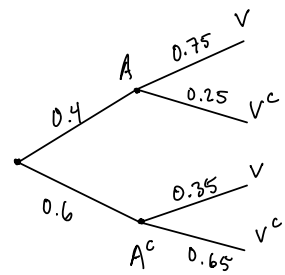
   (b) $P(V \mid A) = 0.75$

   (c) $P(V) = P(V \text{ and } A) + P(V \text{ and } A^c)$

   $= P(V \mid A)\,P(A) + P(V \mid A^c)\,P(A^c)$

   $= (0.75)(0.4) + (0.35)(0.6)$

   $= 0.51$



   (d) $P(A \text{ and } V) = P(V \mid A)\,P(A) = (0.4)(0.75) = 0.3.$

   Since this probability is nonzero, $A \cap V \neq \emptyset$. A, V are not mutually exclusive.

   $P(V \mid A) = 0.75$, different from $P(V) = 0.51$. V, A are not independent.

2. (b), (c) and (e)

3. (a) VI (b) II (c) I (d) V

4. (a) $X \sim \text{Binom}(10, 0.2)$

   (b) $\mu_X = np = (10)(0.2) = 2$

   (c) $P(X=2) = {}_{10}C_2\,(0.2)^2(0.8)^8 = \dfrac{10!}{8!\,2!}(0.04)(0.1678)$

   $= \dfrac{(10)(9)}{2}(0.04)(0.1678) = 0.302$

5. (a) German shepherd dogs

   (b)

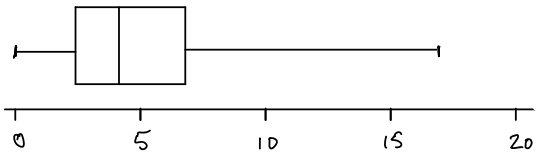
Variables	Type
food	categorical
weight	quantitative

6. (a) unimodal, right-skewed

   (b) $q_1 = 2.5$, $q_3 = 6.5$ (estimated)

   (c) 0, 2.5, 4, 6.5, 17

(d)



(e) $iqr = 6.5 - 2.5 = 4$

$1.5 \times iqr = (1.5)(4) = 6$

values larger than $6.5 + (1.5)\,iqr = 12.5$ are flagged

(f) The mean is larger,

7. (a) $P(X = 7) = 1 - (0.18 + 0.2 + 0.3) = 0.32$

(b) $P(X \geq 5) = P(X=5) + P(X=7) = 0.3 + 0.32 = 0.62$

(c) $\mu_X = (2)(0.18) + (3)(0.2) + (5)(0.3) + (7)(0.32) = 4.7$

8. (a) nrow(dogs)

(b) names(dogs)

(c) mean(~ weight, data = dogs)